

# Market Mood Forecasting

## **Event-Aware, Leakage-Safe Forecasting System (v1.2.0)**

This presentation documents the upgraded Market Mood Forecasting portfolio project, including walk-forward validation, bilingual Gradio deployment, event-aware feature engineering, SHAP explainability, and leakage-safe architecture.

# Technical Overview

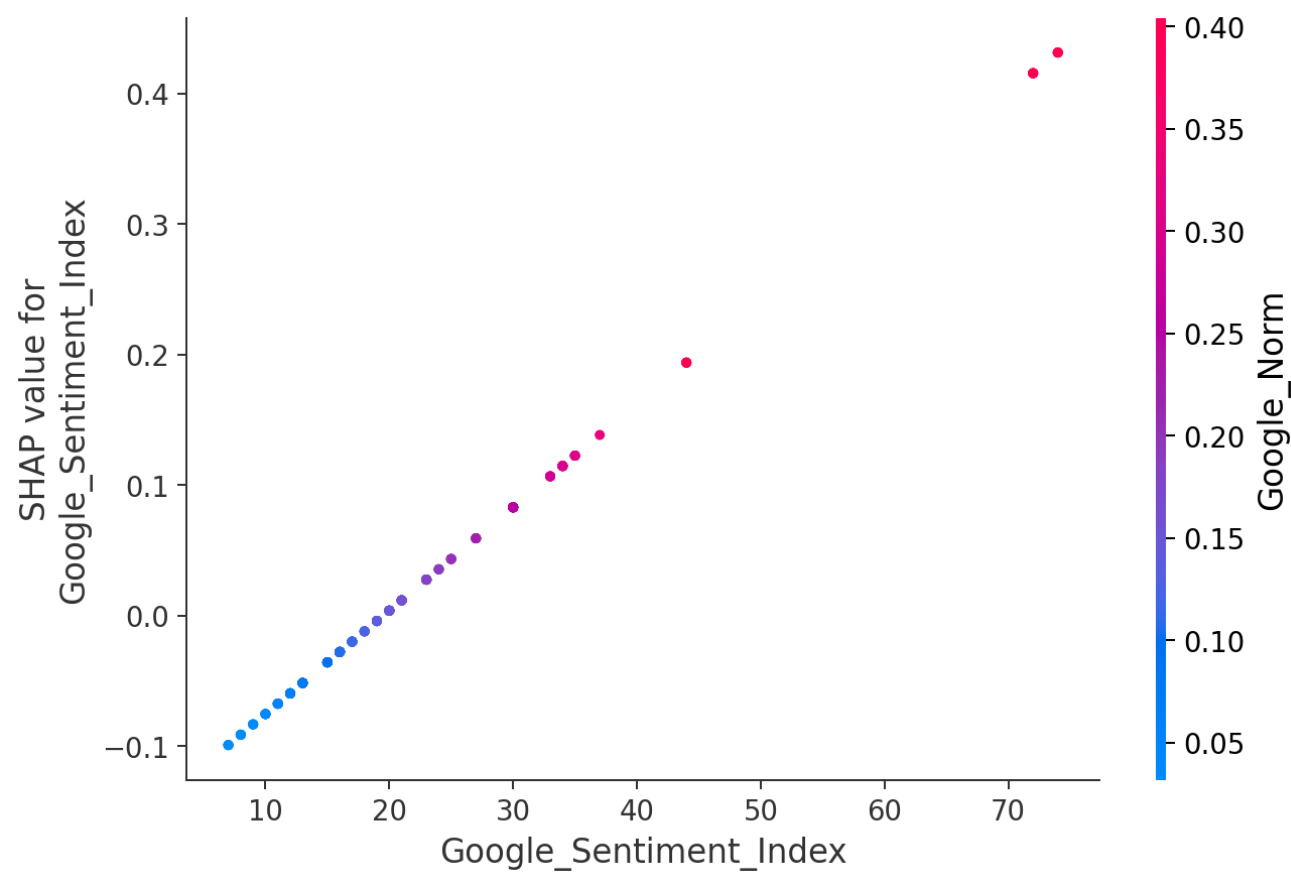
**Objective:** Forecast next-week downside market risk using historical market behavior, volatility signals, sentiment data, and engineered event-risk features.

**Main Upgrades:** Walk-forward validation pipeline Leakage-safe feature engineering Event-risk contextual modeling SHAP explainability integration Bilingual English/German interface Deployment diagnostics panel Demo nudge interaction workflow

| Capability              | Status    |
|-------------------------|-----------|
| Walk-Forward Validation | Completed |
| Event-Risk Features     | Completed |
| SHAP Explainability     | Completed |
| Leakage Guardrails      | Completed |
| Bilingual UX            | Completed |
| Hugging Face Deployment | Completed |

# Project Snapshot 1

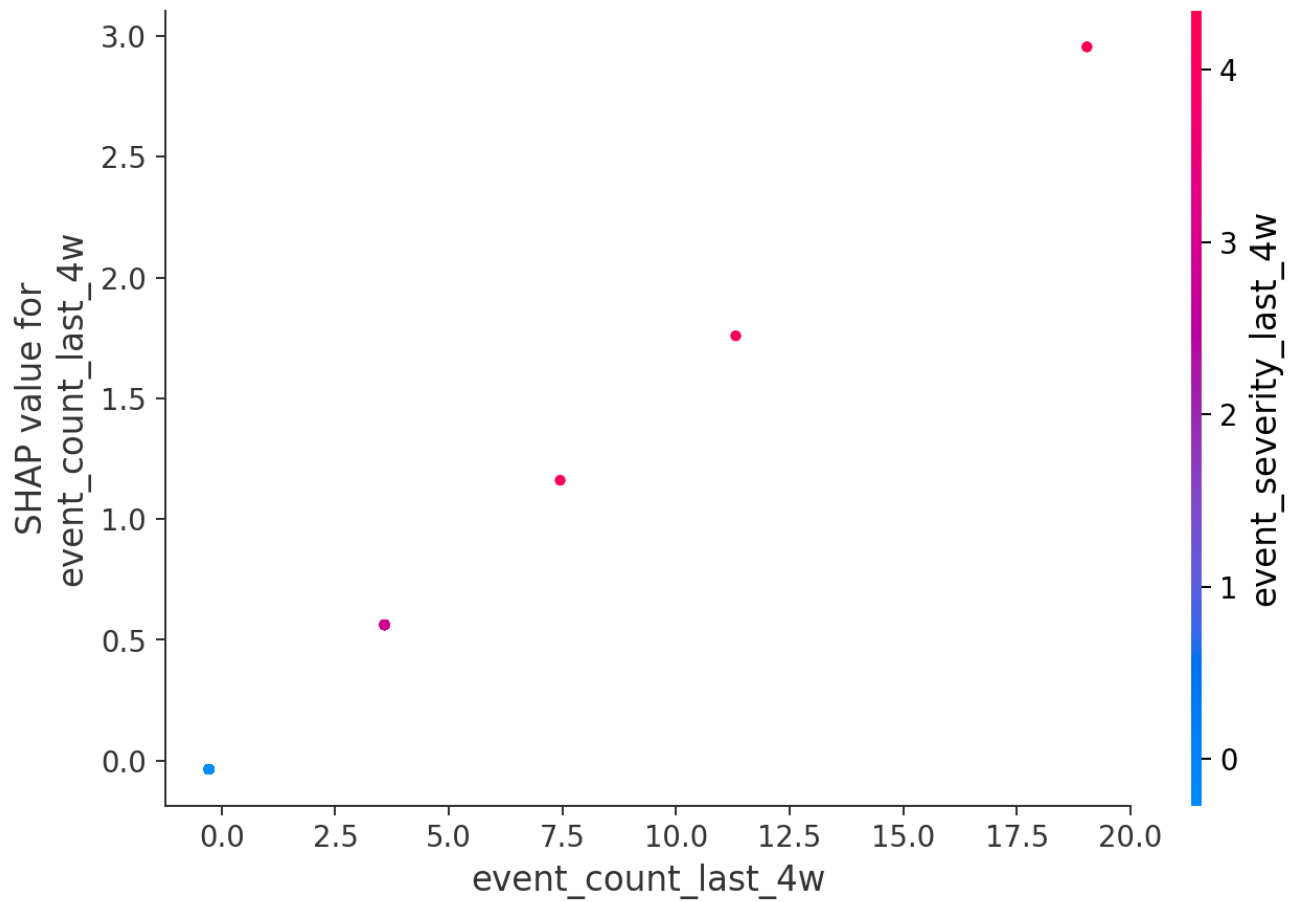
dependence\_Google\_Sentiment\_Index\_v1\_1.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

## Project Snapshot 2

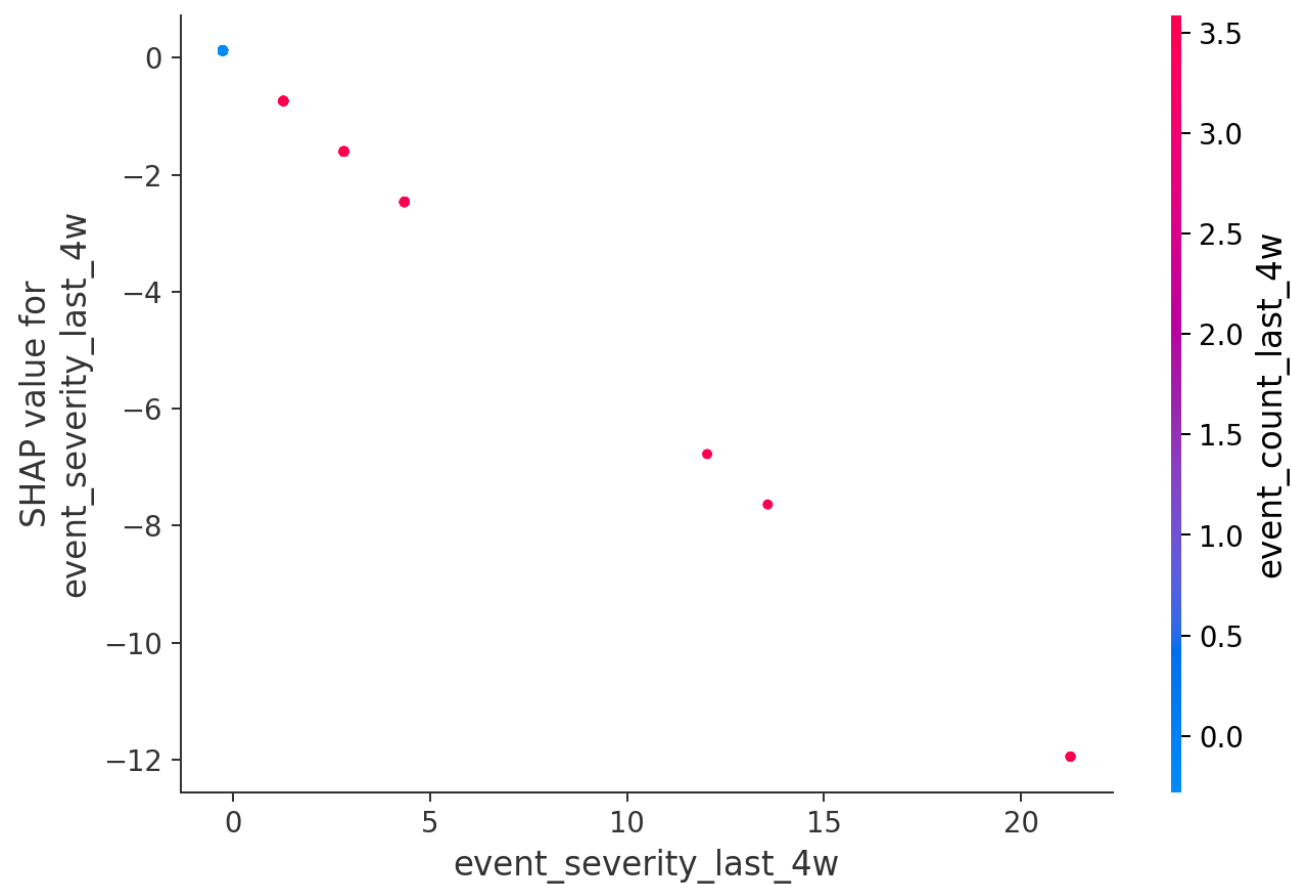
dependence\_event\_count\_last\_4w\_v1\_2\_0.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

# Project Snapshot 3

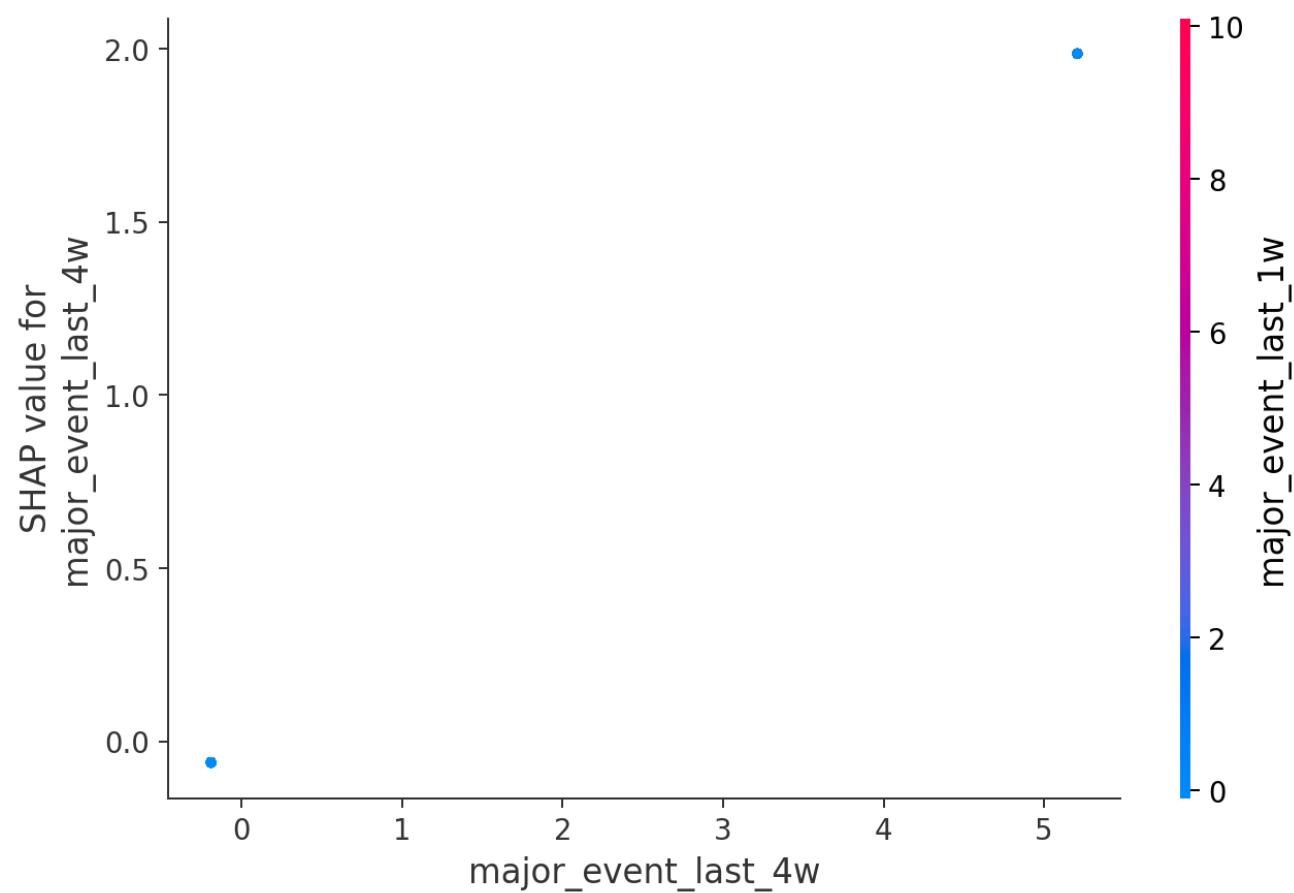
dependence\_event\_severity\_last\_4w\_v1\_2\_0.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

# Project Snapshot 4

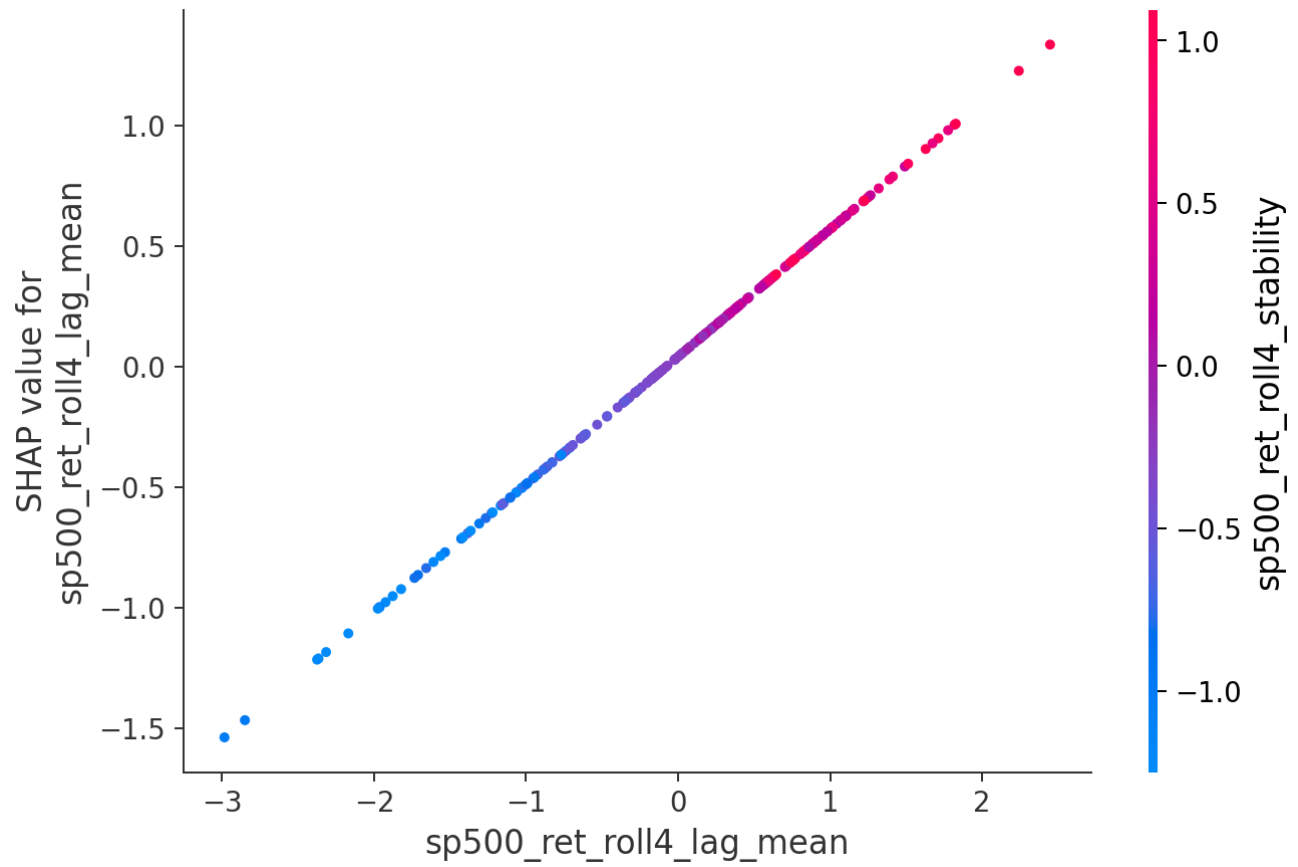
dependence\_major\_event\_last\_4w\_v1\_2\_0.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

## Project Snapshot 5

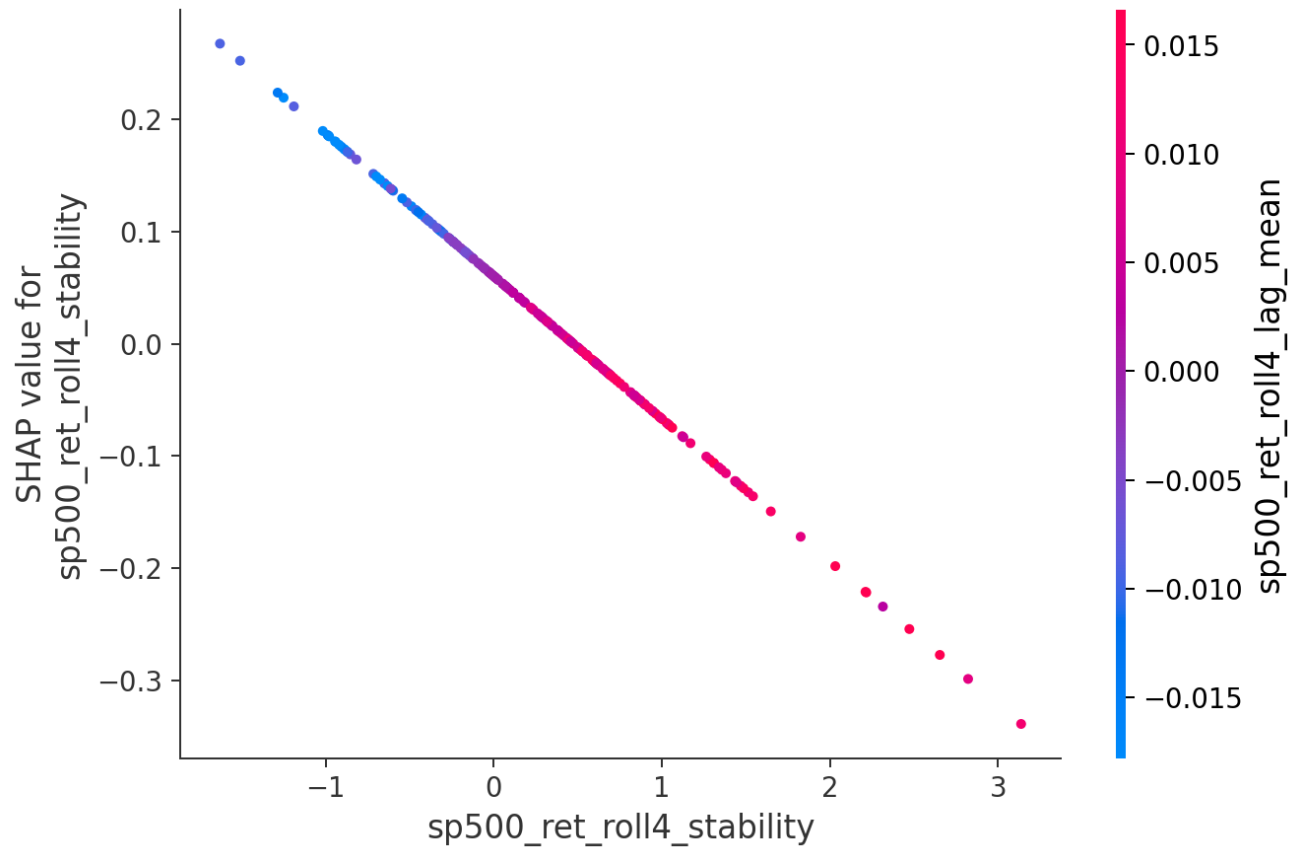
dependence\_sp500\_ret\_roll4\_lag\_mean\_v1\_2\_0.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

## Project Snapshot 6

dependence\_sp500\_ret\_roll4\_stability\_v1\_1.png

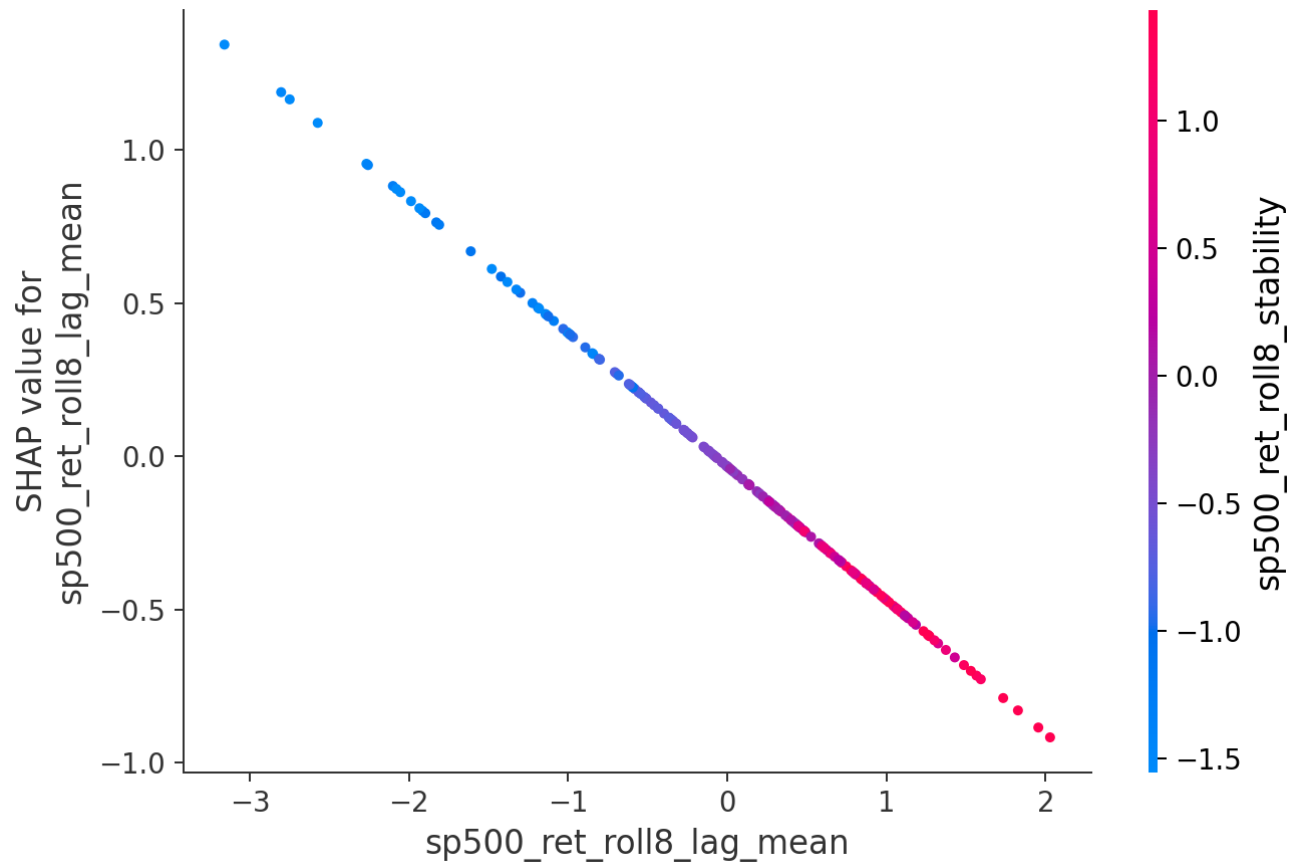


This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.



## Project Snapshot 7

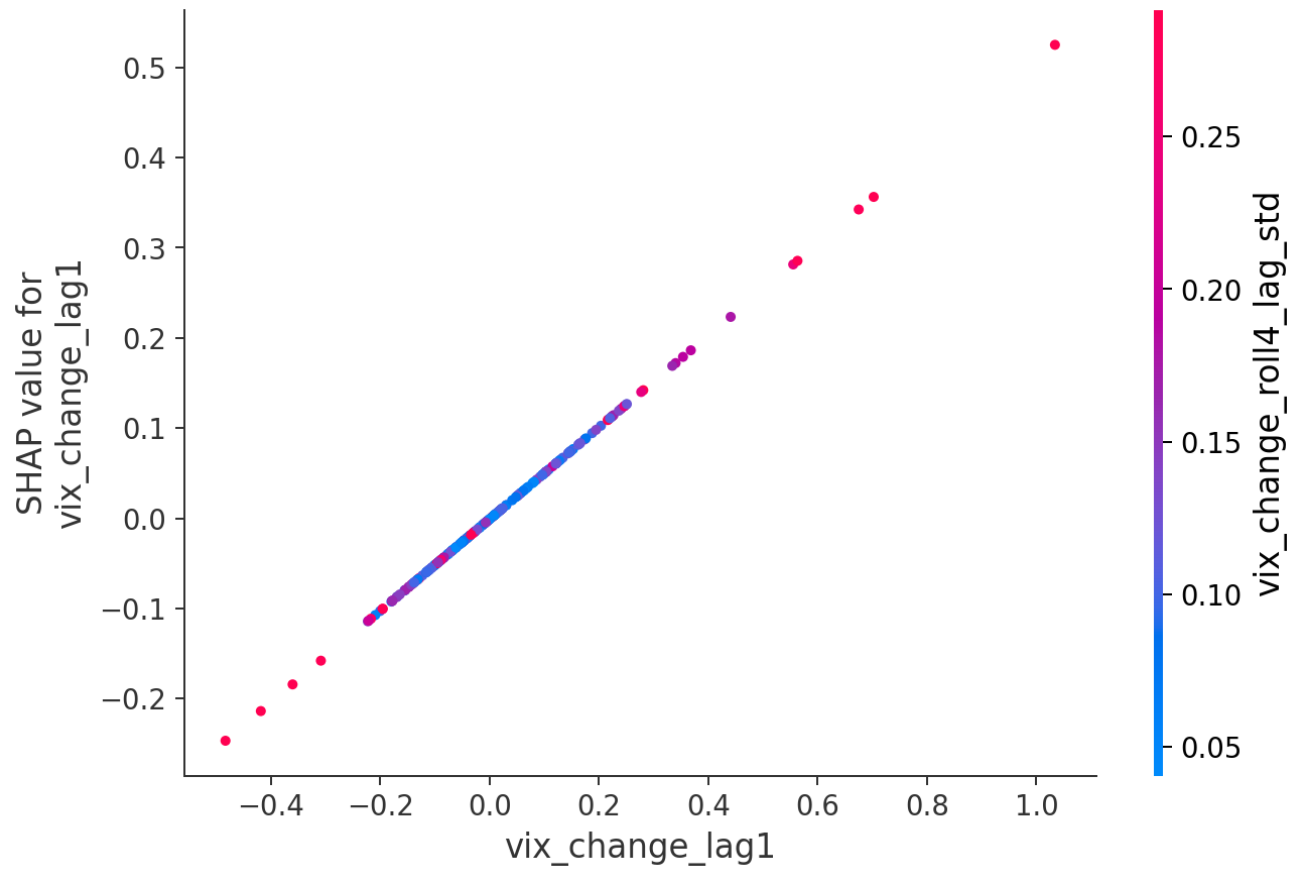
dependence\_sp500\_ret\_roll8\_lag\_mean\_v1\_2\_0.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

## Project Snapshot 8

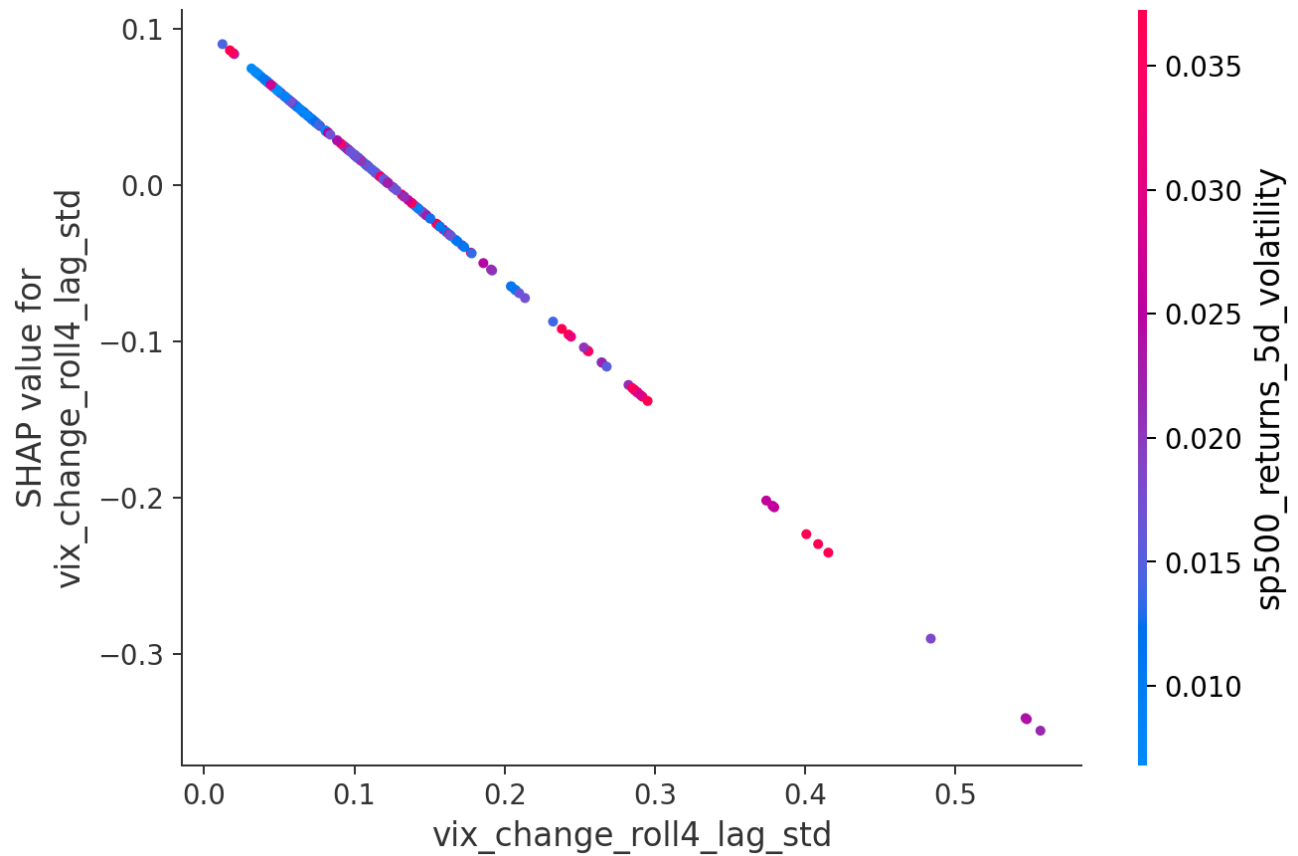
dependence\_vix\_change\_lag1\_v1\_1.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

## Project Snapshot 9

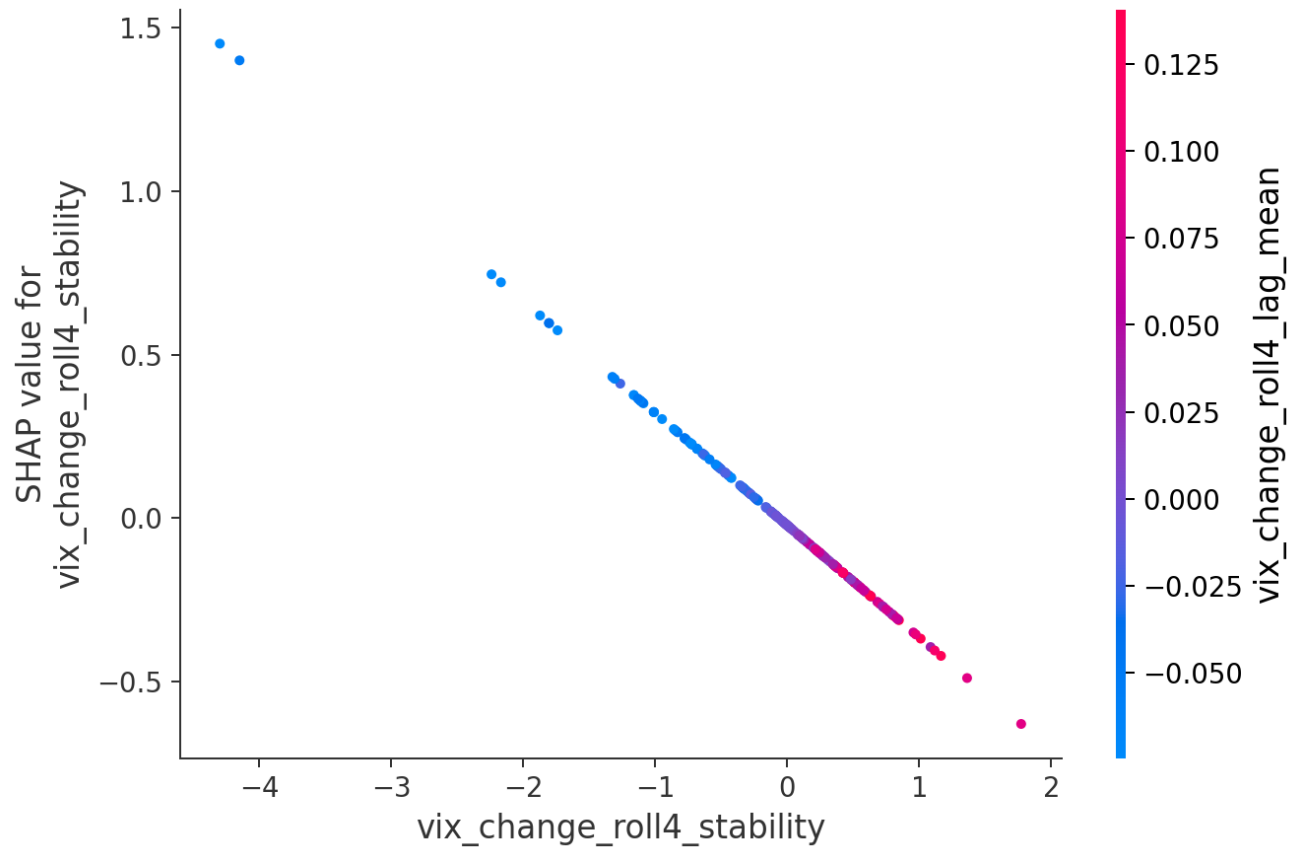
dependence\_vix\_change\_roll4\_lag\_std\_v1\_1.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

## Project Snapshot 10

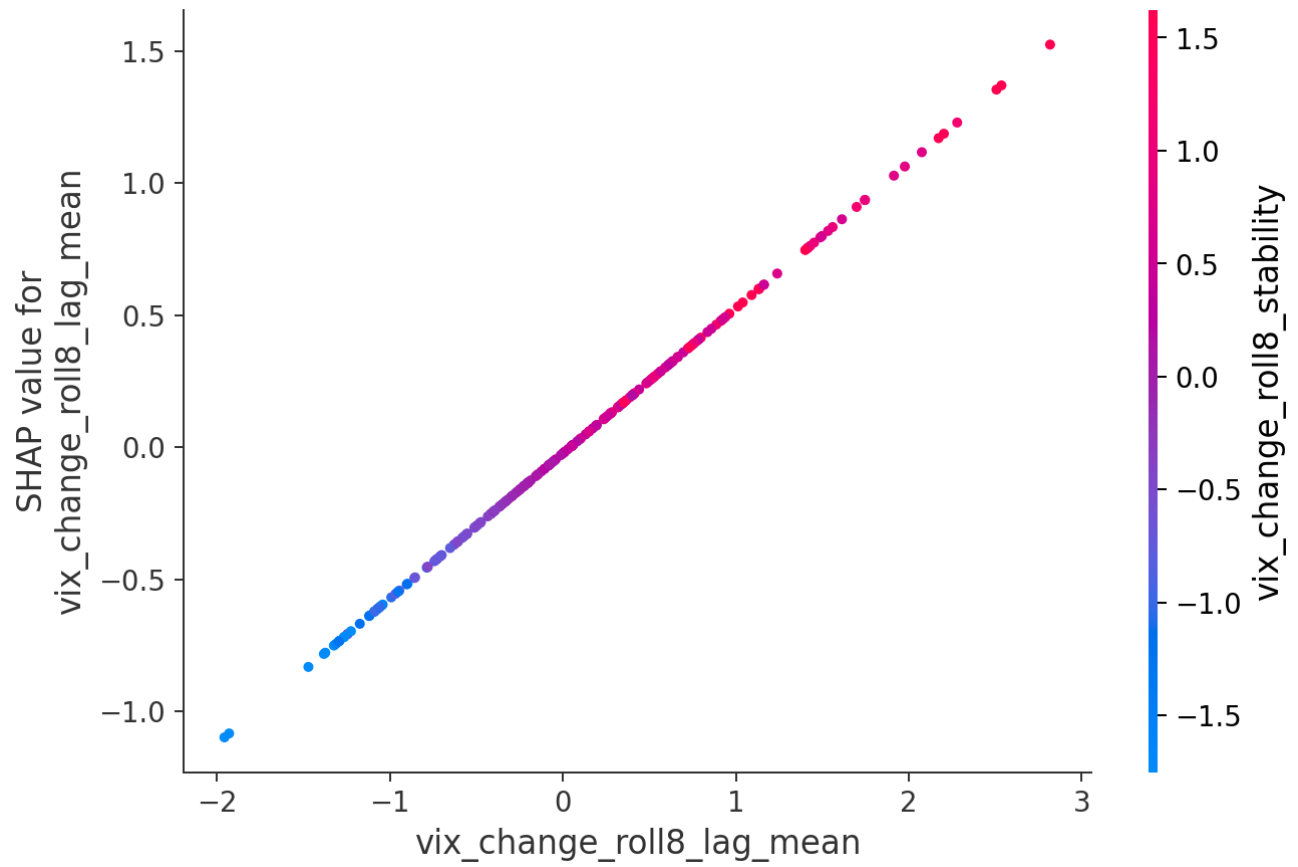
dependence\_vix\_change\_roll4\_stability\_v1\_1.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

## Project Snapshot 11

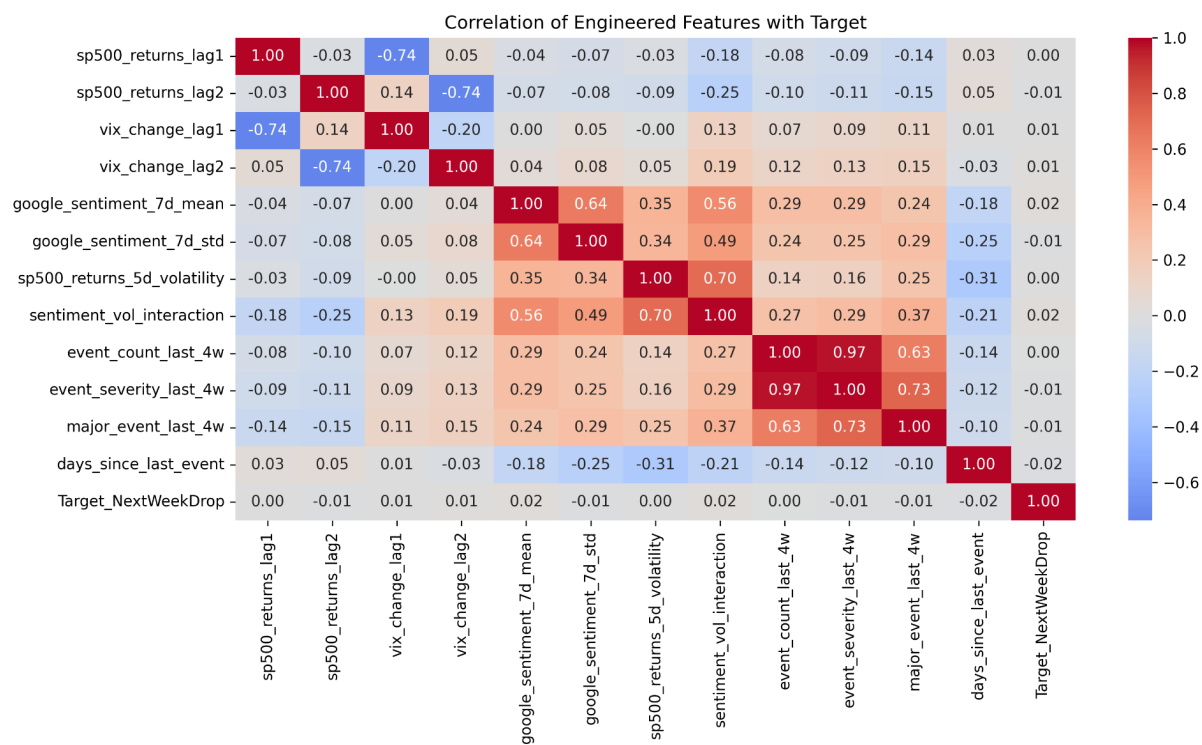
dependence\_vix\_change\_roll8\_lag\_mean\_v1\_2\_0.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

# Project Snapshot 12

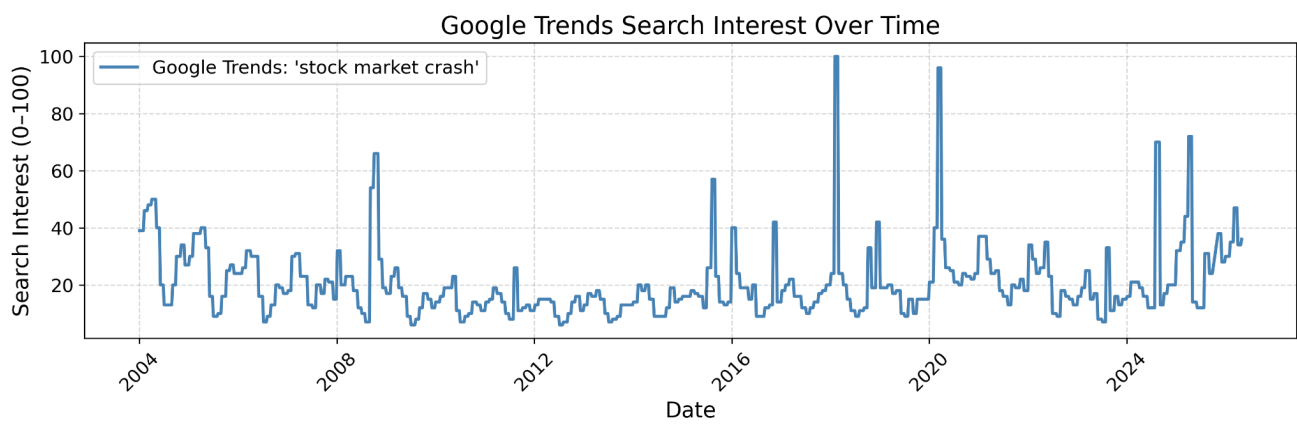
feature\_corr\_heatmap\_v1\_2\_0.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

# Project Snapshot 13

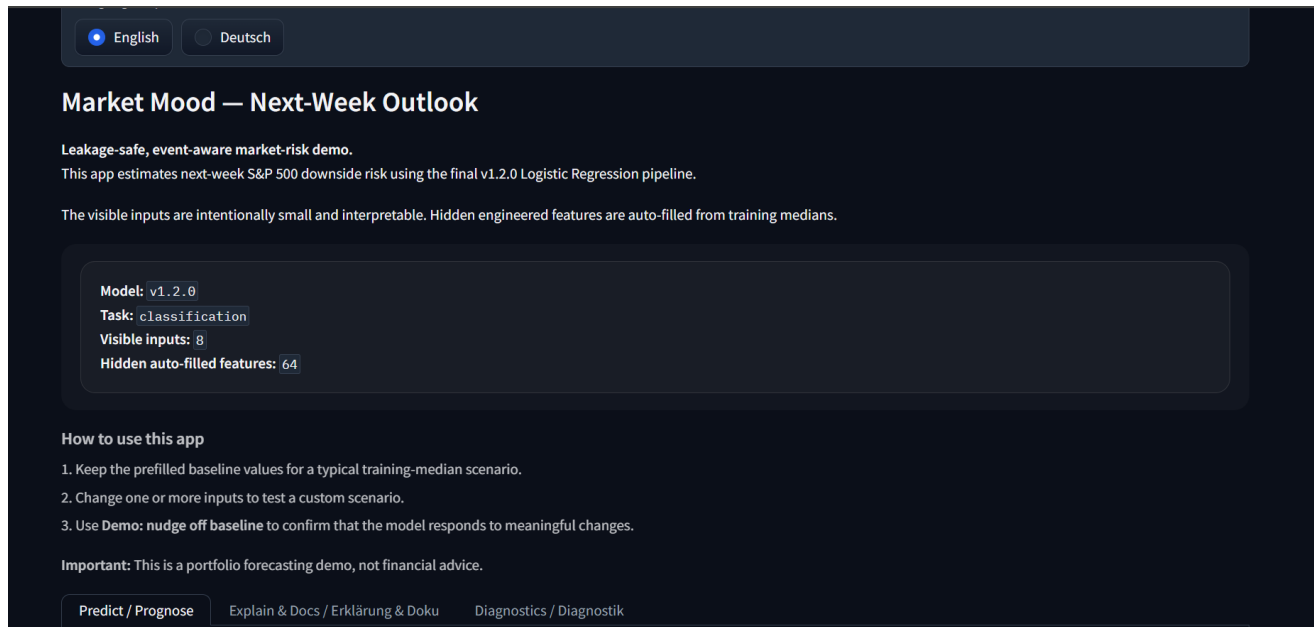
google\_trends\_sentiment.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.

# Project Snapshot 14

image.png



This figure represents part of the upgraded v1.2.0 workflow, including application UX, event-aware feature engineering, time-series visualization, SHAP interpretation, or diagnostics validation.



# Final Interpretation

The final implementation emphasizes methodological correctness, interpretability, deployment reproducibility, and portfolio-quality communication.

**Key Strengths:** Chronological validation methodology Leakage-safe feature architecture Explainable Logistic Regression workflow Event-aware market context modeling Bilingual deployment UX Professional technical documentation

The project demonstrates applied data science skills across: time-series feature engineering market analytics machine learning workflows SHAP explainability deployment integration technical storytelling

*Educational portfolio project only — not investment advice.*